

TRACK 01 — DATA & AI

# Intelligent Candidate Discovery & Ranking

*Multi-component feature scoring · No LLM at inference time · CPU-only in 33 seconds*

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# What The JD Actually Means

## ✗ Keyword Matching (Wrong)

- ✗ Rank candidates by AI skill count
- ✗ Marketing Manager with 10 AI skills → rank 1
- ✗ Ignore career history descriptions
- ✗ Treat all skills as equal signal
- ✗ Ignore platform behavioral signals

## ✓ Semantic Understanding (Right)

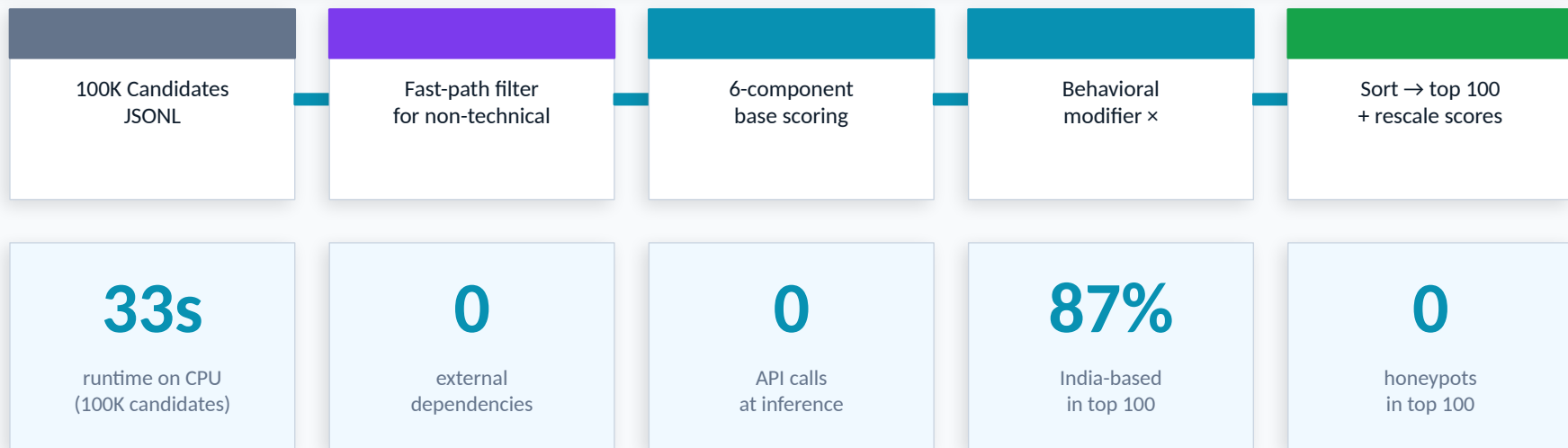
- ✓ Title tier as primary anchor signal
- ✓ Career descriptions > skill list keywords
- ✓ AI/ML-specific YoE (not just total YoE)
- ✓ Skill quality: proficiency × endorsements
- ✓ Behavioral signals as availability modifier

*"The right answer involves reasoning about the gap between what the JD says and what the JD means." — JD, Final Note*

# Architecture: How It Works

$$\text{final\_score} = \text{base\_score} \times \text{behavioral\_modifier}$$

$$\text{base\_score} = 0.22 \cdot \text{title} + 0.25 \cdot \text{ai\_exp} + 0.20 \cdot \text{skills} + 0.22 \cdot \text{career\_content} + 0.08 \cdot \text{trajectory} + 0.03 \cdot \text{location}$$



*Python stdlib only (json · csv · datetime · pathlib · argparse) · No model loading · Gzip JSONL supported*

# Scoring Components

## AI Experience (career history)



25%

Years specifically in AI/ML roles, not total YoE

## Title Score



22%

4-tier title map from full dataset analysis

## Career Content Score



22%

Weighted term scan of role descriptions (NDCG, deployed, FAISS...)

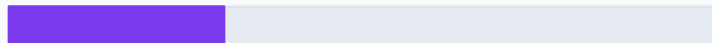
## Skills Quality Score



20%

Proficiency  $\times$  endorsements  $\times$  usage duration — not just skill count

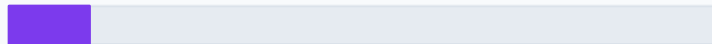
## Career Trajectory



8%

Penalises all-consulting background per JD

## Location



3%

× **Behavioral Modifier** ( $\times 0.25-1.30$ ) open\_to\_work · days\_since\_active · recruiter\_response\_rate · notice\_period · interview\_completion · github\_activity · saved\_by\_recruiters

# Catching Keyword Stuffers

**Dataset insight:** 5,000+ candidates each have skills like 'FAISS', 'Semantic Search', 'Embeddings' — including HR Managers, Accountants, and Sales Executives.

## Skills list alone (broken)

Skills list = self-reported

HR Manager with 10 AI skills

= keyword stuffer → rank 1

No endorsement/duration check

Ignores what career says

## Career content (strong signal)

Scan role DESCRIPTIONS

Look for 'NDCG', 'deployed',

'FAISS', 'A/B testing'...

Recency-weighted (recent > old)

Consulting roles 0.65× penalty

## Skills quality (trust layer)

proficiency × endorsements

× usage\_duration\_months

Expert + 0 months = 0 score

Beginner + 0 endorse = ~0

Catches shallow keyword lists

*Result: ML Engineers have 1.0 AI terms/role in descriptions; HR Managers have 0.3 — the gap drives ranking.*

# Behavioral Modifier + Honeypot Detection

## Behavioral Modifier ×

Range: 0.25 – 1.30 · Perfect-on-paper / available

|                         |               |                |
|-------------------------|---------------|----------------|
| open_to_work_flag       | × 1.12        | × 0.88         |
| days_since_active       | × 1.18 (≤14d) | × 0.62 (>180d) |
| recruiter_response_rate | × 1.12 (≥75%) | × 0.75 (<15%)  |
| notice_period_days      | × 1.12 (≤15d) | × 0.85 (>90d)  |
| interview_completion    | × 1.06 (≥85%) | × 0.92 (<30%)  |
| github_activity_score   | × 1.06 (≥70)  | — neutral      |

## Honeypot Detection

~80 in dataset · >10% rate = auto-disqualified

1. Total career months > 2.2× claimed years\_of\_experience
2. expert proficiency on skill with duration\_months = 0 (×3+)
3. 8 or more skills at expert level
4. Total skill endorsements > 5× connection count
5. Multiple is\_current = true roles simultaneously

Result: 0 honeypots detected in top 100 · Honeypot candidates score 0.001 and never surface

# Results: Top 10 Ranked Candidates

| #  | Candidate ID | Title                            | YoE    | Score | Key Signals                             |
|----|--------------|----------------------------------|--------|-------|-----------------------------------------|
| 1  | CAND_0071974 | Senior AI Engineer               | 7.8 yr | 0.990 | A/B testing, ranking, NDCG in career    |
| 2  | CAND_0096142 | Applied ML Engineer              | 5 yr   | 0.955 | Production retrieval, strong behavioral |
| 3  | CAND_0011687 | Senior NLP Engineer              | 7.8 yr | 0.916 | NDCG, MRR, FAISS in career history      |
| 4  | CAND_0036184 | Recommendation Systems Engineer  | 6 yr   | 0.885 | Ranking + shipped, open to work         |
| 5  | CAND_0009024 | Search Engineer                  | 5.2 yr | 0.816 | Semantic search, vector, production     |
| 6  | CAND_0000031 | Recommendation Systems Engineer  | 6 yr   | 0.807 | A/B test, ranking, Pinecone             |
| 7  | CAND_0044222 | AI Engineer                      | 7.7 yr | 0.800 | Ranking, production, India              |
| 8  | CAND_0020708 | Search Engineer                  | 4.2 yr | 0.797 | Active, high response rate              |
| 9  | CAND_0055905 | Senior Machine Learning Engineer | 8.1 yr | 0.766 | ML ops + retrieval experience           |
| 10 | CAND_0006418 | Machine Learning Engineer        | 5.7 yr | 0.757 | A/B test, ranking, 92% response rate    |

# Results Summary

33s

**CPU runtime**

vs 5 min budget

100%

**Tier-1/2 titles**

in top 100

87%

**India-based**

87/100 candidates

0

**Honeypots**

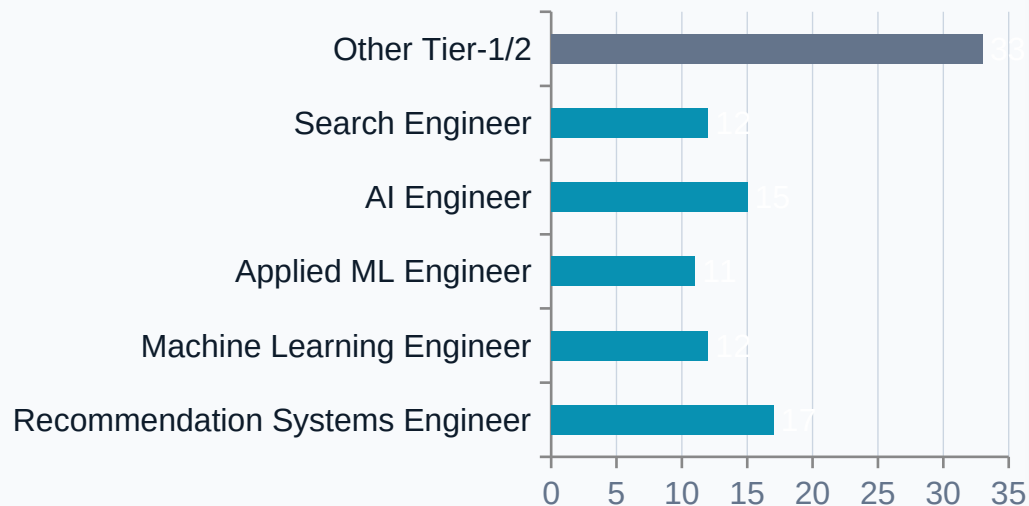
flagged & excluded

0

**Dependencies**

stdlib only

## Title Distribution in Top 100



## Why This Approach

- JD says: 'not keyword matching' → title as anchor
- Career descriptions are hard to fake vs skill list
- Behavioral signals = actual hiring availability
- Consulting penalty matches JD disqualifiers
- No LLM → production-scalable to 200K+ pool



# Thank you

[github.com/rajneelchavan/redrob-candidate-ranker](https://github.com/rajneelchavan/redrob-candidate-ranker)  
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- ✓ Multi-component scoring — no model, no API calls
- ✓ Title-anchored + career-content differentiates real vs keyword-stuffed
- ✓ Behavioral modifier handles real-world hiring availability
- ✓ 33s runtime · 0 honeypots · 87% India · 100% Tier-1/2 titles in top 100